ModelIt!

Student Activity Pack

Can We Reverse Deforestation Before It's Too Late?

Modeling Carbon Sinks, Tipping Points, and Reforestation Economics

Grade 11L2 | NGSS HS-LS2-4, HS-ESS3-3 | Stability and Change / Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do you think tropical forests are called the 'lungs of the planet'?

2 How much of the world's original forest do you think has been cut down, and why?

3 Draw a diagram showing how you think forests affect rainfall in their region.

4 If we planted enough trees, could we solve climate change? Why or why not?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Deforestation Rate | Forest Carbon Stock | Evapotranspiration Rate | Regional Rainfall | Fire Susceptibility | Reforestation Rate

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Current Trajectory

10M hectares/year deforestation, no reforestation

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Reforestation Matching

Equal deforestation and reforestation rates

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Amazon Tipping Point

Amazon loses 25%+ of forest cover

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Biodiversity Index — The richness and abundance of species within the forest, which depends on intact habitat connectivity — fragmented forests lose species through edge effects and isolation even if total area is unchanged

Soil Carbon Release — When forest is cleared, soil organic carbon accumulated over centuries is exposed to air and decomposed, releasing massive quantities of CO_2 — tropical forest soils store 40-60% of total ecosystem carbon

Agricultural Commodity Price — Global market prices for beef, soy, palm oil, and timber that drive economic incentives for deforestation — when commodity prices rise, deforestation accelerates regardless of conservation policies

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If deforestation releases billions of tons of carbon and pushes forests toward dieback tipping points, can planting trees fast enough actually reverse the damage — or are we just buying time?
Use evidence from your simulation to support your answers.

1. **Why does your model show that reforestation alone cannot solve the problem if deforestation continues at current rates?**

2. **How does the evapotranspiration feedback loop make tropical deforestation qualitatively different from temperate deforestation?**

3. **What does your model reveal about the critical difference between preventing deforestation and replanting after deforestation?**

4. **How would economic incentives need to change to make standing forest more valuable than cleared land in your model?**

5. **What does the tipping point in your model suggest about how much time we have to act before irreversible savannification begins?**

8 STEM Challenge: Design a Forest Conservation and Restoration Strategy

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YOUR ENGINEERING MISSION

Design an integrated strategy for a tropical region that combines deforestation prevention, strategic reforestation, and economic incentives to keep the forest system above its tipping point.

THE SCENARIO

A tropical country with 70% of its original forest remaining has committed to zero net deforestation by 2035. Current deforestation is driven by cattle ranching and soy farming. Your team must design a strategy that protects remaining forest, restores degraded areas, and creates economic alternatives that make standing forest more valuable than cleared land.

Engineering Design Thinking

1. HOW WOULD YOU USE SATELLITE MONITORING TO DETECT AND RESPOND TO ILLEGAL DEFORESTATION IN REAL TIME?

2. WHAT MIX OF NATIVE SPECIES AND PLANTING PATTERNS WOULD CREATE GENUINELY RESTORED FOREST RATHER THAN MONOCULTURE PLANTATIONS?

3. HOW WOULD YOU STRUCTURE CARBON CREDIT PROGRAMS OR PAYMENTS FOR ECOSYSTEM SERVICES TO COMPENSATE LANDOWNERS FOR NOT CLEARING FOREST?

4. HOW WOULD YOU PHASE YOUR STRATEGY TO ACHIEVE THE GREATEST IMPACT IN THE FIRST FIVE YEARS WHILE BUILDING TOWARD LONG-TERM SUSTAINABILITY?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Forest Ecologists and Conservation Scientists study forest ecosystem dynamics, design restoration strategies, and monitor deforestation using satellite data. They work for organizations like the World Wildlife Fund, The Nature Conservancy, national forest agencies, and research universities, earning $60,000–$120,000/year. Carbon Market Analysts specializing in forest-based carbon credits earn $75,000–$140,000/year.